



wazuh.

SIEM



SURICATA®

Software-Based Firewall Integration with Wazuh SIEM

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1. Task Summary

Objective:

Implement a firewall rule on Kali Linux using IPSet to block all incoming traffic from Sweden-based IP addresses, monitor this activity using Wazuh SIEM, and validate the effectiveness of the block with alert logs.

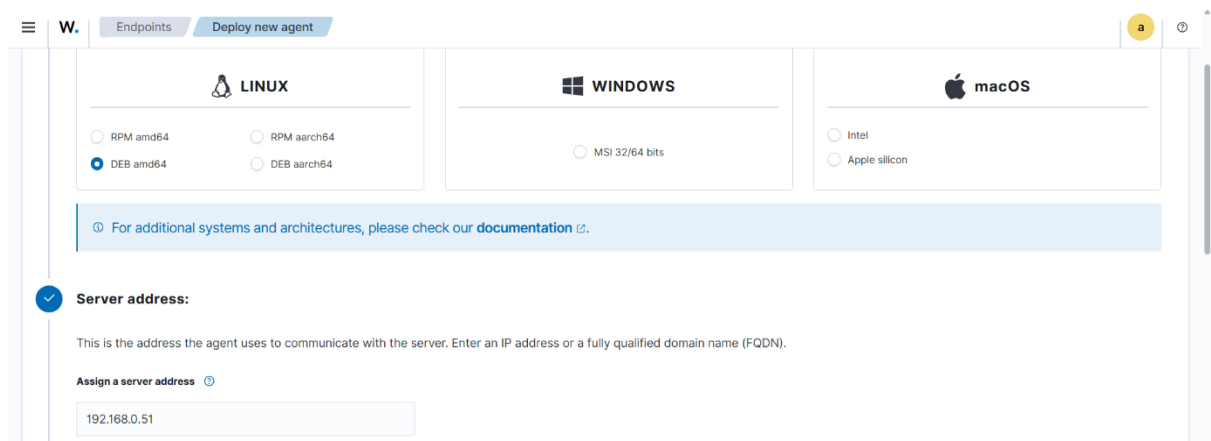
2. Tools and Setup

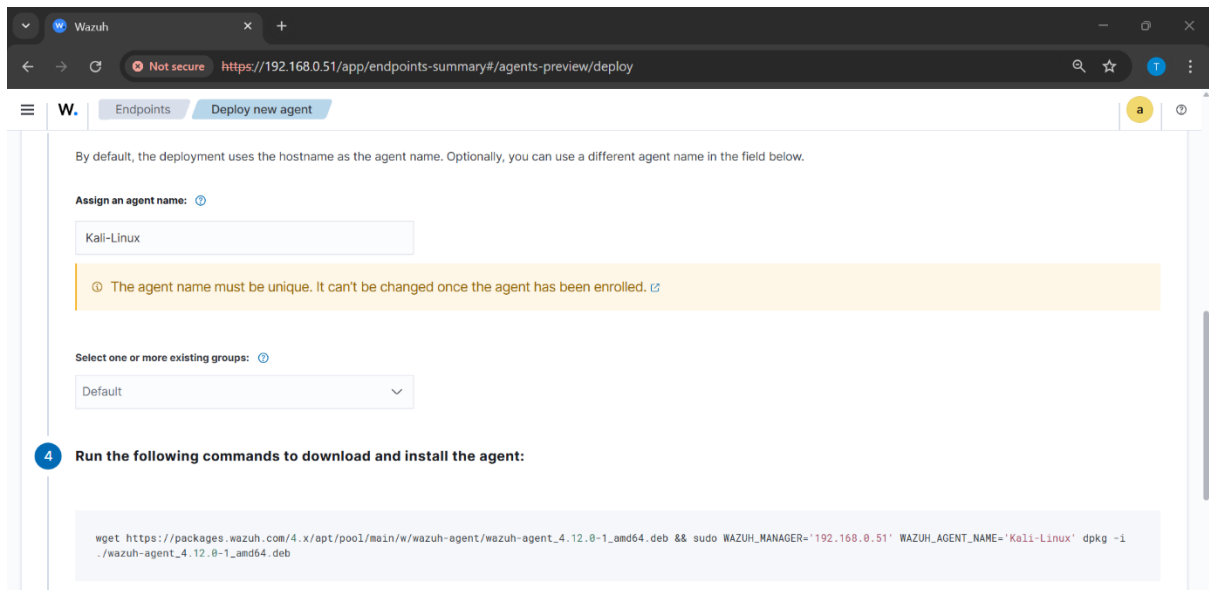
Lab Environment

Component	Description
OS	Kali Linux (VirtualBox)
Firewall Tool	IPSet + IPTables
Monitoring Tool	Wazuh Agent + Wazuh Manager
Target IP Source	IPDeny Sweden IP List
Network Mode	Bridged Adapter

3. Step-by-Step Implementation:

A. Deploy Wazuh-agent into Kali Linux

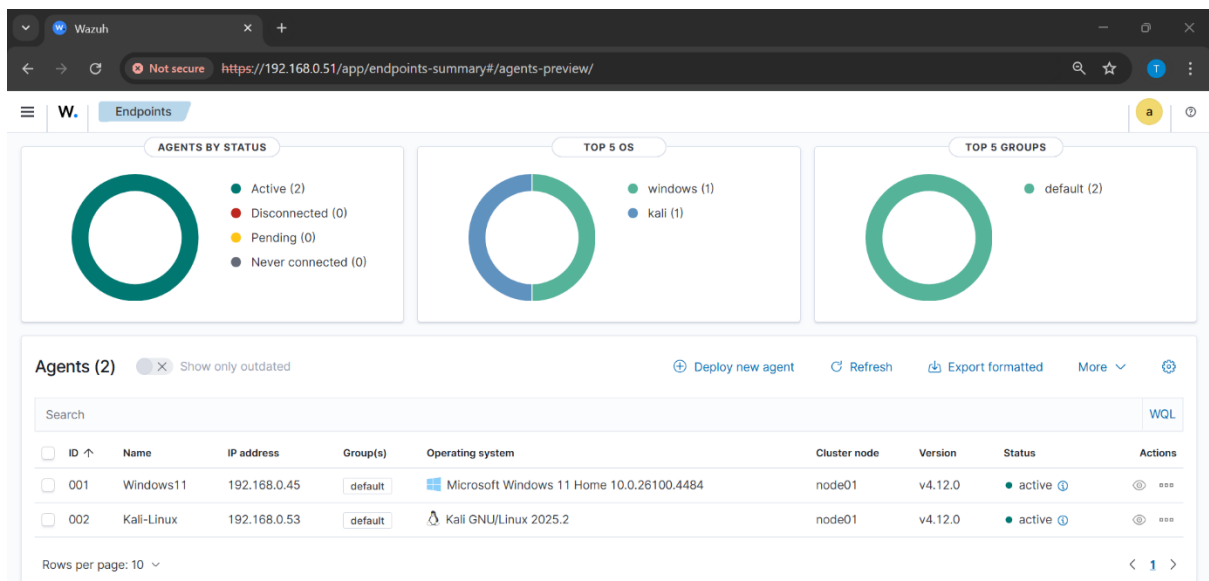




B. Copy Paste the Commands into Kali Linux Terminal:



Agent has been connected properly:



Step 2: Now Install Suricata into Kali Linux

A. Update Kali & Install Suricata:

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali ~
(kali@kali)~$ sudo apt install suricata -y
Error: Unable to locate package suricata
(kali@kali)~$ sudo apt update
sudo apt install suricata -y
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.4 MB]
90% [3 Contents-amd64 48.7 MB/51.4 MB 95%] 1,735 kB/s 2s
```

B. Check the info & Version:

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali ~
(kali@kali)~$ suricata --build-info
This is Suricata version 7.0.10 RELEASE
Features: NFQ PCAP SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HTTP2_DECOMPRESSION_HA
VE LUA HAVE_JA3 HAVE_JA4 HAVE_LUAJIT HAVE_LIBJANSSON TLS TLS_C11 MAGIC RUST POPCNT64
SIMD support: SSE_4_2 SSE_4_1 SSE_3 SSE_2
Atomic intrinsics: 1 2 4 8 16 byte(s)
64-bits, Little-endian architecture
GCC version 14.2.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v0.5.50, linked against LibHTP v0.5.50
```

C. Check your Kali OS IP:

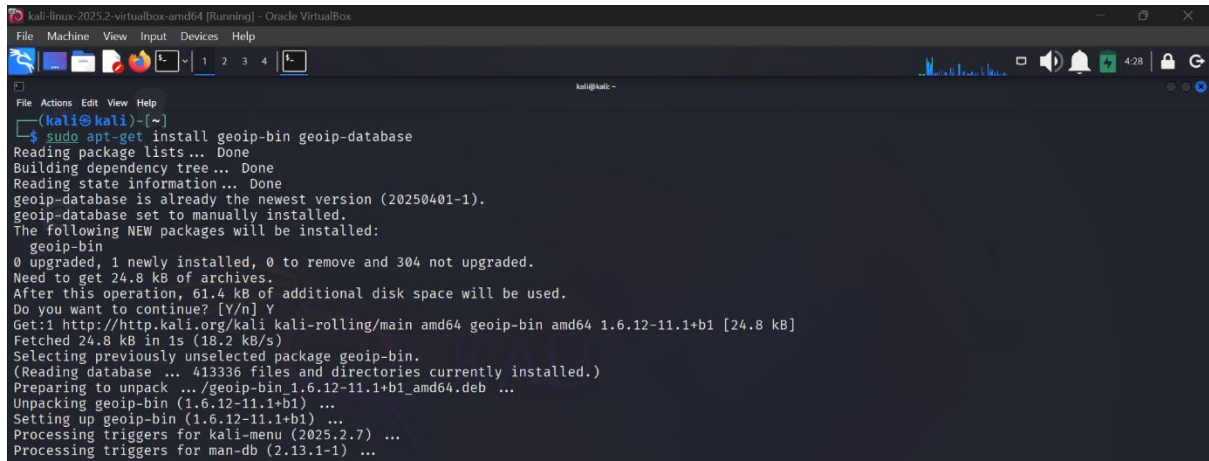
```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali ~
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.53/24 brd 192.168.0.255 scope global dynamic noprefixroute eth0
        valid_lft 85556sec preferred_lft 85556sec
    inet6 fe80::42e9:98c6:1914:39d8/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

D. Add you ip addr:

```
vars:
# more specific is better for alert accuracy and performance
address-groups:
  HOME_NET: "[192.168.0.53/24]"
  #HOME_NET: "[192.168.0.0/16]"
  #HOME_NET: "[10.0.0.0/8]"
  #HOME_NET: "[172.16.0.0/12]"
  #HOME_NET: "any"

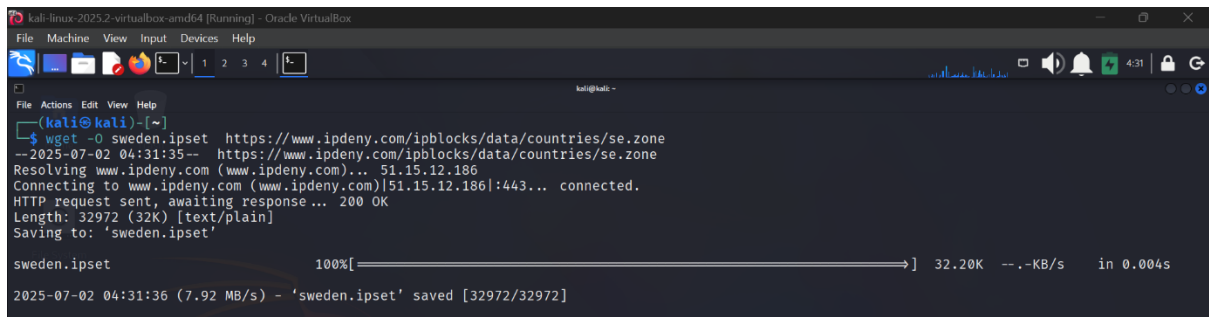
  EXTERNAL_NET: "!$HOME_NET"
  #EXTERNAL_NET: "any"
```

E. Installing GeoIP Tools for Country-Based Traffic Filtering on Kali Linux:



```
kali@kali:~$ sudo apt-get install geoip-bin geoip-database
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
geoip-database is already the newest version (20250401-1).
geoip-database set to manually installed.
The following NEW packages will be installed:
  geoip-bin
0 upgraded, 1 newly installed, 0 to remove and 304 not upgraded.
Need to get 24.8 kB of archives.
After this operation, 61.4 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 geoip-bin amd64 1.6.12-11.1+b1 [24.8 kB]
Fetched 24.8 kB in 1s (18.2 kB/s)
Selecting previously unselected package geoip-bin.
(Reading database ... 41336 files and directories currently installed.)
Preparing to unpack .../geoip-bin_1.6.12-11.1+b1_amd64.deb ...
Unpacking geoip-bin (1.6.12-11.1+b1) ...
Setting up geoip-bin (1.6.12-11.1+b1) ...
Processing triggers for kali-menu (2025.2.7) ...
Processing triggers for man-db (2.13.1-1) ...
```

F. Now Download Sweden.ip set:

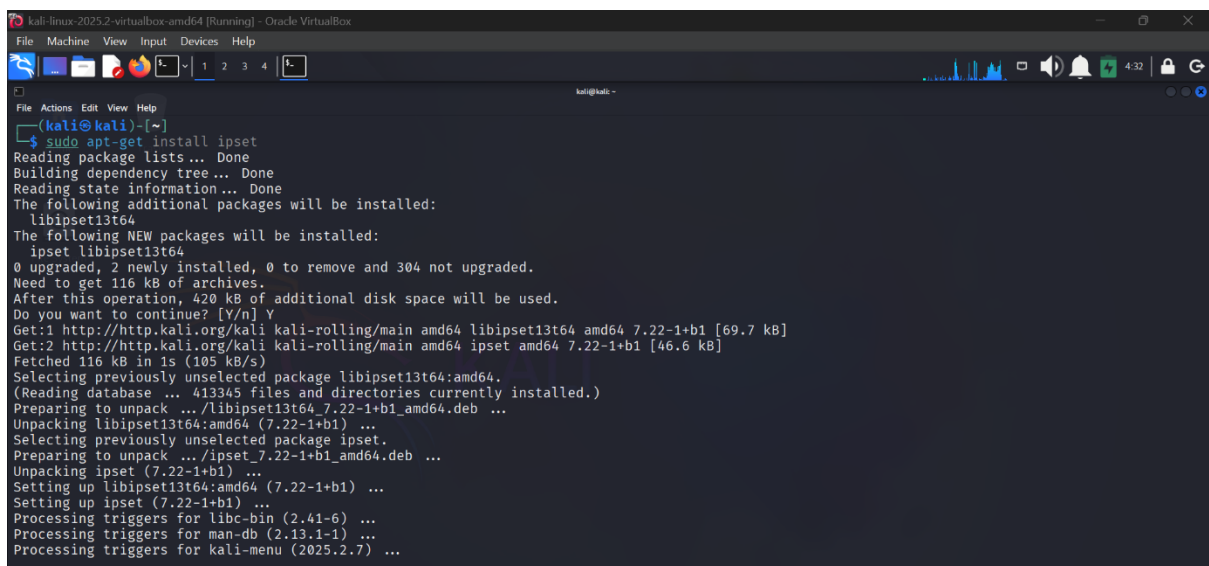


```
kali@kali:~$ wget -O sweden.ipset https://www.ipdeny.com/ipblocks/data/countries/se.zone
--2025-07-02 04:31:35-- https://www.ipdeny.com/ipblocks/data/countries/se.zone
Resolving www.ipdeny.com (www.ipdeny.com)... 51.15.12.186
Connecting to www.ipdeny.com (www.ipdeny.com)[51.15.12.186]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 32972 (32K) [text/plain]
Saving to: 'sweden.ipset'

sweden.ipset          100%[=====] 32.20K  --.-KB/s  in 0.004s

2025-07-02 04:31:36 (7.92 MB/s) - 'sweden.ipset' saved [32972/32972]
```

G: Setting Up IPSet for Geo-Based Blocking:



```
kali@kali:~$ sudo apt-get install ipset
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libipset13t64
The following NEW packages will be installed:
  ipset libipset13t64
0 upgraded, 2 newly installed, 0 to remove and 304 not upgraded.
Need to get 116 kB of archives.
After this operation, 420 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libipset13t64 amd64 7.22-1+b1 [69.7 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 ipset amd64 7.22-1+b1 [46.6 kB]
Fetched 116 kB in 1s (105 kB/s)
Selecting previously unselected package libipset13t64:amd64.
(Reading database ... 41345 files and directories currently installed.)
Preparing to unpack .../libipset13t64_7.22-1+b1_amd64.deb ...
Unpacking libipset13t64:amd64 (7.22-1+b1) ...
Selecting previously unselected package ipset.
Preparing to unpack .../ipset_7.22-1+b1_amd64.deb ...
Unpacking ipset (7.22-1+b1) ...
Setting up libipset13t64:amd64 (7.22-1+b1) ...
Setting up ipset (7.22-1+b1) ...
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.2.7) ...
```

Step 3: Add IPs to IPSet from File Verify IPSet

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali ~
$ while IFS= read -r ip; do sudo ipset add sweden_ips "$ip"; done < sweden.ipset
(kali@kali)~$ sudo ipset list sweden_ips
Name: sweden_ips
Type: hash:net
Revision: 7
Header: family inet hashsize 1024 maxelem 65536 bucketsize 12 initval 0xe4c25fec
Size in memory: 63720
References: 0
Number of entries: 2117
Members:
185.130.244.0/22
192.245.97.0/24
185.122.72.0/23
5.44.192.0/20
202.163.0.0/19
185.84.240.0/22
194.0.246.0/24
185.142.228.0/22
31.209.0.0/18
31.216.224.0/21
78.40.40.0/21
185.103.208.0/22
185.170.180.0/22
185.176.28.0/22
192.71.0.0/16
185.209.196.0/22
213.89.128.0/17
91.199.68.0/24
46.227.72.0/21
```

Step 5: Block Traffic from Sweden

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali ~
$ sudo iptables -I INPUT -m set --match-set sweden_ips src -j DROP
(kali@kali)~$ sudo iptables -I OUTPUT -m set --match-set sweden_ips dst -j DROP
(kali@kali)~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- any any anywhere anywhere match-set sweden_ips src

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
0 0 DROP all -- any any anywhere anywhere match-set sweden_ips dst
```

Step 6: Adding Rules

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: /home/kali
root@kali ~# sudo cat /etc/suricata/rules/ipset.rules
alert ip any any -> any any (msg:"ALERT: Sweden traffic spotted!"; content:"SE"; sid:1000002; rev:1;)

root@kali ~# sudo nano /etc/suricata/rules/local.rules
root@kali ~# sudo cat /etc/suricata/rules/local.rules
drop http any any -> any any (msg:"DROP Instagram, Facebook, YouTube access"; content:"instagram.com"; http_host; sid:1000005; rev:1;)
drop http any any -> any any (msg:"DROP Instagram, Facebook, YouTube access"; content:"facebook.com"; http_host; sid:1000006; rev:1;)
drop http any any -> any any (msg:"DROP Instagram, Facebook, YouTube access"; content:"youtube.com"; http_host; sid:1000007; rev:1;)
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1; rev:1;)
```

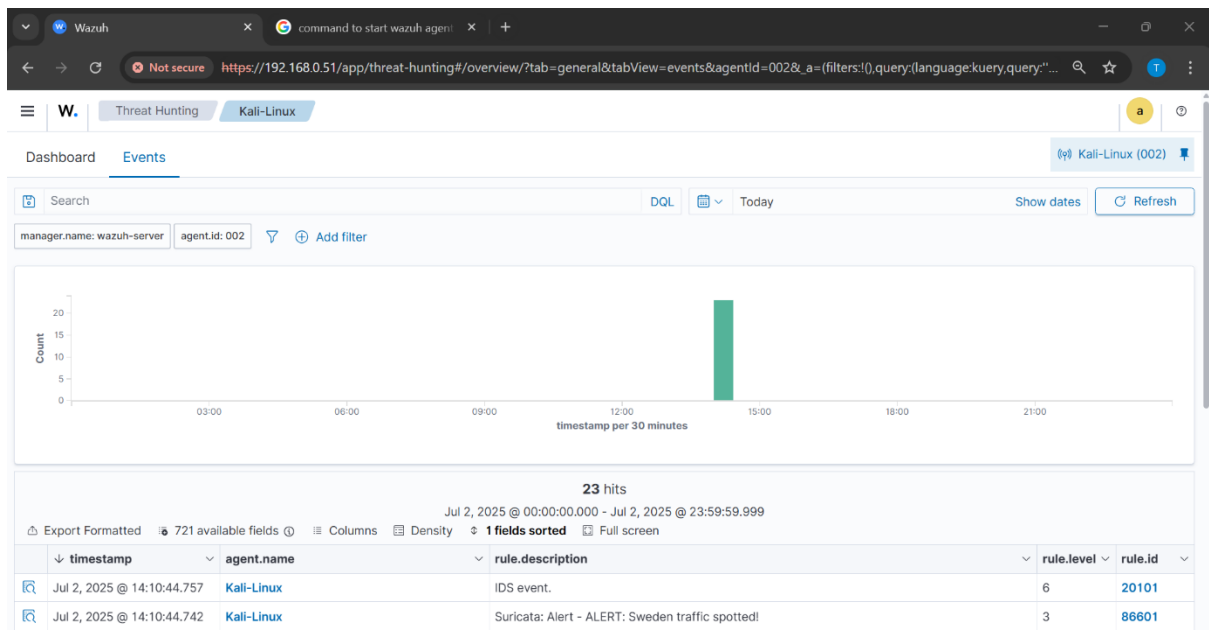
Step 7: Configuring OSSEC to monitor Suricata logs

```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: /home/kali
root@kali ~# nano /etc/suricata/suricata.yaml
GNU nano 8.4 /etc/suricata/suricata.yaml *
#
# See Nmaptech NTPL documentation other hashmodes and details on their use.
#
# This parameter has no effect if auto-config is disabled.
#
hashmode: hashstuplesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##
default-rule-path: /var/lib/suricata/rules
rule-files:
- /etc/suricata/rules/ipset.rules
- /etc/suricata/rules/local.rules

##
## Auxiliary configuration files.
##
```

Step 8: Monitor in Wazuh Dashboard



Wazuh Threat Hunting Kali-Linux

Jul 2, 2025 @ 00:00:00.000 - Jul 2, 2025 @ 23:59:59.999

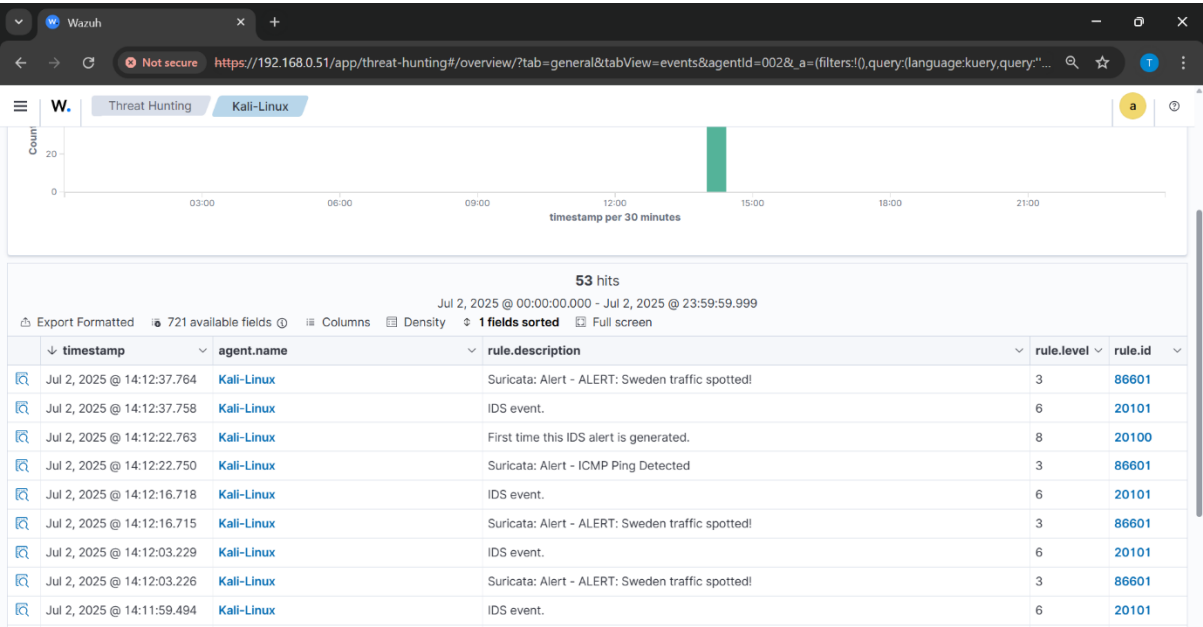
Export Formatted 721 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 2, 2025 @ 14:10:44.757	Kali-Linux	IDS event.	6	20101
Jul 2, 2025 @ 14:10:44.742	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:10:29.731	Kali-Linux	IDS event.	6	20101
Jul 2, 2025 @ 14:10:29.672	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:10:22.849	Kali-Linux	IDS event.	6	20101
Jul 2, 2025 @ 14:10:22.803	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:09:52.191	Kali-Linux	IDS event.	6	20101
Jul 2, 2025 @ 14:09:52.189	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:09:50.102	Kali-Linux	First time this IDS alert is generated.	8	20100
Jul 2, 2025 @ 14:09:50.102	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:09:40.185	Kali-Linux	First time this IDS alert is generated.	8	20100
Jul 2, 2025 @ 14:09:40.176	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:09:38.721	Kali-Linux	First time this IDS alert is generated.	8	20100
Jul 2, 2025 @ 14:09:38.720	Kali-Linux	Suricata: Alert - ALERT: Sweden traffic spotted!	3	86601
Jul 2, 2025 @ 14:09:35.952	Kali-Linux	First time this IDS alert is generated.	8	20100

Using Ping command to verify the logs:

```
kali-linux-20752-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@kali:~/home/kali
# sudo systemctl start wazuh-agent
# systemctl start wazuh-agent
# systemctl restart wazuh-agent
# ping facebook.com
PING facebook.com (157.240.227.35) 56(84) bytes of data:
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=1 ttl=54 time=30.5 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=2 ttl=54 time=40.5 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=3 ttl=54 time=18.2 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=4 ttl=54 time=36.1 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=5 ttl=54 time=28.7 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=6 ttl=54 time=24.3 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=7 ttl=54 time=24.7 ms
64 bytes from edge-star-mini-shv-01-mct1.facebook.com (157.240.227.35): icmp_seq=8 ttl=54 time=21.6 ms
```

Here are logs after ping to Facebook:



Validation Tests:

Test Scenario	Result
Ping/traceroute from Sweden IP	Blocked/Dropped
Alert visibility in Wazuh	Successful
IPSet integrity verified	Successful

Observations & Findings:

- Sweden IPs successfully blocked at firewall level using IPSet.
- Suricata/Wazuh Agent can be extended to monitor those IPs.
- Wazuh dashboard reflected system-level alerts (if logs monitored).
- Highly efficient method for mass-country IP blocking with low resource usage.

Conclusion:

Successfully implemented IPSet-based blocking for Sweden IP ranges and validated the rule through traffic blocking and alert monitoring in Wazuh. This method provides lightweight country-based control with full SOC visibility.